FEDERAL SPECIFICATION

GRINDING AND POLISHING MACHINE, DENTAL LABORATORY, FLOOR MOUNTED, 110 VOLT, 60 CYCLE, A. C.

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a floor mounted dental laboratory grinding and polishing machine, suitable for operation on 115 volt, 60 cycle, alternating current.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CC–M–636—Motors, Fractional Horsepower (Alternating-Current).

NN–P–530—Plywood; Flat-Panel.

QQ–A–359—Aluminum Alloy Plate and Sheet 3003.

QQ–W–461—Wire, Steel (Carbon), Bare and Coated.

UU–P–271—Paper, Wrapping, Waterproofed Kraft.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–C–11456—Crates; Wood, Nailed, Unsheathed, Domestic Shipment, 1500 Pounds Maximum Load.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each bidder shall submit to the contracting agency proof that the equipment he proposes to supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the equipment conforms to this requirement.

3.1.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his equipment conforms to the published standards including methods of tests of the Underwriters' Laboratories, Inc.

3.2 Materials.—Materials shall conform in all respects to the specifications listed in section 2, and as hereinafter specified. When materials are used which are not specifically designated they shall be of high commercial quality throughout, and suitable in all respects for the purpose intended. The materials in all cases shall be consistent with the strength required for safety and durability. Machines in their entirety shall be of such design and material as to assure rigid, rugged, and durable construction, and shall perform in a satisfactory manner the functions for which such machines are intended.

3.2.1 *Sheet aluminum.*—The sheet aluminum shall be alloy 3003, in conformance with Federal Specification QQ–A–359 for H14 temper. Thickness of sheet shall be 0.032 inch (No. 20 B.&S.).

3.2.2 *Plywood.*—The plywood shall be 5-ply, either Douglas fir, interior type, grade B-D, or hardwood, medium density, type II, grade 2, in conformance with Federal Specification NN–P–530.

3.2.3 *Galvanized wire.*—The wire used in the fabrication of the screen as specified in 3.4.10 shall be zinc-coated steel wire conforming to Federal Specification QQ–W–461, except the wire shall be 0.035 ± 0.003 inch in diameter.

3.3 Style and design.—Style and design of the grinding and polishing machine shall be as shown on figure 2.

3.4 Components.—The grinding and polishing machine shall consist essentially of a cabinet with a removable front panel, tabletop, hoods, splash pans, work shelf, blower unit, polishing lathe, filters, tray, screen, and accessories as hereinafter specified.

3.4.1 *Cabinet.*—The sides, back, bottom, and removable front panel of the cabinet shall be fabricated of sheet aluminum alloy as specified in 3.2.1, with all parts securely welded. The sides, back, and bottom shall be riveted to an angle iron framework. The vertical framework forming the four corners shall extend below the bottom of the cabinet, serving as legs. Bearing plates shall be welded to the end of the legs. The front panel of the cabinet shall be equipped with suitable means for easy removal, without the use of tools. The outside measurements of the cabinet shall be 28½ ± ½ inches in width, 17 ± 1 inches in depth, and 38 ± ½ inches in height when measured from the floor to the working surface of the tabletop.

3.4.2 *Tabletop.*—Tabletop shall be made of ⅝-inch plywood as specified in 3.2.2, bounded on sides and back with a solid wood railing, and covered on top with sheet aluminum as specified in 3.2.1. The railing shall be 1¾ inches high and ¾ inch thick. The working surface of the tabletop shall be 29½ ± ½ inches wide and 17½ ± ½ inches deep.

3.4.3 *Hoods and splash pans.*—Hoods and splash pans shall be fabricated of sheet aluminum as specified in 3.2.1. Hoods shall be 13 inches high, 7 inches wide, and 13 inches deep, and shall be screwed to the tabletop. Each hood shall be equipped with a standard lampholder. Splash pans shall be separate from the hoods for convenient removal and

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cleaning, and suitable means shall be provided for holding the pans securely in place. One pan shall have an aluminum wire screen welded to back section, which shall extend 6¼ inches above top of pan, and shall be bent forward at an angle of 15 degrees. The other pan shall have a solid aluminum sheet, as specified in 3.2.1, welded to back section, which shall extend 3⅝ inches above top of pan, and shall be bent forward at an angle of 30 degrees. Splash pans shall be 10½ inches long, 7 inches wide, and 2 inches high. All edges and corners of the hoods and splash pans shall be rounded, ground smooth, or otherwise treated to eliminate burrs and sharp or ragged edges. All seams shall be smoothly and continuously soldered, and thoroughly wiped to remove excess solder and flux. As an alternate, heliarc welding may be employed.

3.4.4 *Work shelf.* — Work shelf shall be made of ⅝-inch plywood as specified in 3.2.2, and shall consist of the working surface with a railing ⅝ inch high nailed to top and surrounding the front and sides. Work shelf shall be supported by two iron rods, securely attached to the tabletop and the under surface of the shelf. Shelf shall be 30½ inches wide and 8½ inches in length, and shall be bent forward at an angle of approximately 15 degrees. Height of shelf above tabletop shall be 17⅜ inches.

3.4.5 *Blower unit.*—Blower housing, wheel, and motor shall be assembled as one unit, and shall be mounted on a spring chassis to eliminate vibration. Spring chassis shall be securely bolted to the framework of the cabinet. Blower shall be 8 inches in diameter and made entirely of aluminum. Blower housing shall be made of steel. Blower shall have a capacity of 900 cubic feet per minute (c.f.m.).

3.4.5.1 *Motor.*—Motor shall conform to the applicable requirements of Federal Specification CC–M–636 for single-phase, capacitor-start type. Motor shall be ¼ horsepower designed for operation on 115 volt, 60 cycle, alternating current. Motor shall have a single operating speed of 1725 revolutions per minute (r.p.m.) and shall be provided with fully enclosed sealed-for-life ball bearings.

3.4.6 *Polishing lathe.*—Unless otherwise specified, a polishing lathe shall be furnished (see 6.1). Motor for polishing lathe shall conform to the applicable requirements of Federal Specification CC–M–636 for single-phase, split-phase type. Motor shall be ¼ or ⅓ horsepower designed for operation on 115 volt, 60 cycle, alternating current. Motor shall have a single operating speed of 3450 r.p.m., and shall be provided with fully enclosed sealed-for-life ball bearings. Height from base of motor to shaft shall be 6 inches. Polishing lathe shall be securely bolted to the tabletop.

3.4.7 *Connecting cords.*—Connecting cords for motors shall consist of a rubber-covered service cable with an indestructible type rubber connection plug. The current rating of the cords shall be not less than that of the motor. External connecting cord shall be not less than 8 feet in length.

3.4.8 *Filters.*—Two fiber-glass filters shall be positioned 10 inches above the bottom of the cabinet. Each filter shall measure 25 inches wide, 14 ± 2 inches long, and 2 inches thick. Filters shall be easily removed for cleaning.

3.4.9 *Tray.*—A removable tray shall be located at the bottom of the cabinet. Tray shall be 1¼ inches high and of sufficient length and width to accommodate the filters specified in 3.4.8. Tray shall be fabricated of sheet aluminum as specified in 3.2.1.

3.4.10 *Screen.* — A galvanized-steel wire screen as specified in 3.2.3, woven four to the inch in both directions, shall be located directly below the lower filter.

3.4.11 *Tolerances.*—Commercial tolerances shall apply for gage size or thickness of material. Unless otherwise specified, a plus or

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minus tolerance of 1/4 inch shall apply on all dimensional requirements.

3.4.12 *Accessories.*—Four standard steel taper chucks shall be supplied with each unit, two right, and two left. Over-all length of chucks shall be approximately 4 1/4 inches.

3.5 **Finish.**—The framework, blower housing, and supports for the work shelf shall be finished with a suitable gray paint. The wooden portions of the grinding and polishing machine shall have a coat of clear varnish.

3.6 **Identification marking.** — Each lathe shall have a corrosion-resistant metal nameplate permanently affixed thereto. The plate shall bear the model or style designation, serial number, current, voltage, cycle, and name or registered trade-mark of the manufacturer. The markings on the plate shall be clear and permanent, and shall be accomplished by die stamping into the surface of the metal. Alternate methods of marking the nameplate (other than decalcomania) may be submitted to the contracting officer for approval.

3.6.1 *Identification marking for Military Medical Supply Agency procurement.*—In addition to the marking required in 3.6, the nameplate shall be marked with the medical services symbol conforming to figure 1.

3.7 **Service data.**—The contractor shall enclose with each unit two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number.

3.7.1 *For Military Medical Supply Agency procurement.*—The contractor shall mail six additional copies of the service data required in 3.7, to the contracting officer for distribution.

3.8 **Workmanship.**—Workmanship shall be first class throughout. The grinding and polishing machines shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 **Inspection.**—The inspector shall ascertain that the grinding and polishing machine has been examined and tested in accordance with specification requirements.

4.3 **Tests.**—Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

4.4 **Supplier responsibilities for inspection.** —The supplier shall be required to perform examinations and tests to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government to be used for executing tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

Levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102 (for civil agencies only).

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5.1 Packaging.

5.1.1 *Unit of issue.*—One grinding and polishing machine, as specified, constitutes one unit.

5.1.2 *Level A.*

5.1.2.1 *Unit preparation.*—Each unit shall be disassembled as far as is practicable. Surfaces subject to corrosion shall be cleaned and treated with a suitable rust preventive. All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching or damage to the unit.

5.1.3 *Level C.*—Unit shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*—Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with Federal Specification PPP–B–601, table II, using type III, class 1 plywood or PPP–B–621, table IV. Closure and strapping shall comply with appendix of applicable specification. Each box shall be lined with a waterproof barrier fabricated of material conforming to Federal Specification UU–P–271 and sealed with a suitable waterproof adhesive. The exterior container design shall include three skids of not less than 1½ inches in thickness applied crosswise of base, so positioned as to facilitate handling with a forklift truck, and to maintain unit in an upright position. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, blocks or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

5.2.2 *Level B.*—Each unit shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP–B–601, table I, using type III, class 1 plywood, PPP–B–621, table II, or Military Specification MIL–C–11456. The exterior container design shall include three skids of not less than 1½ inches in thickness applied crosswise of base, so positioned as to facilitate handling with forklift truck, and to maintain unit in an upright position. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, blocks or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.

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(*b*) When polishing lathe is not required (see 3.4.6).

(*c*) When polishing lathe only is required (see 3.4.6).

(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item appearing in the Armed Services Medical Stock List:

6520–538–9250 GRINDING AND POLISHING MACHINE, DENTAL LABORATORY, Floor Mounted, 110 volt, 60 cycle, A.C.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 Transportation description.

6.4.1 Transportation description applicable to this item is:

Dental Machines

Carload Minimum Weight 20,000 pounds. Subject to Rule 34, Uniform Freight Classification.

Motor Volume Minimum Weight 20,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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FIGURE 1.—Medical services symbol.

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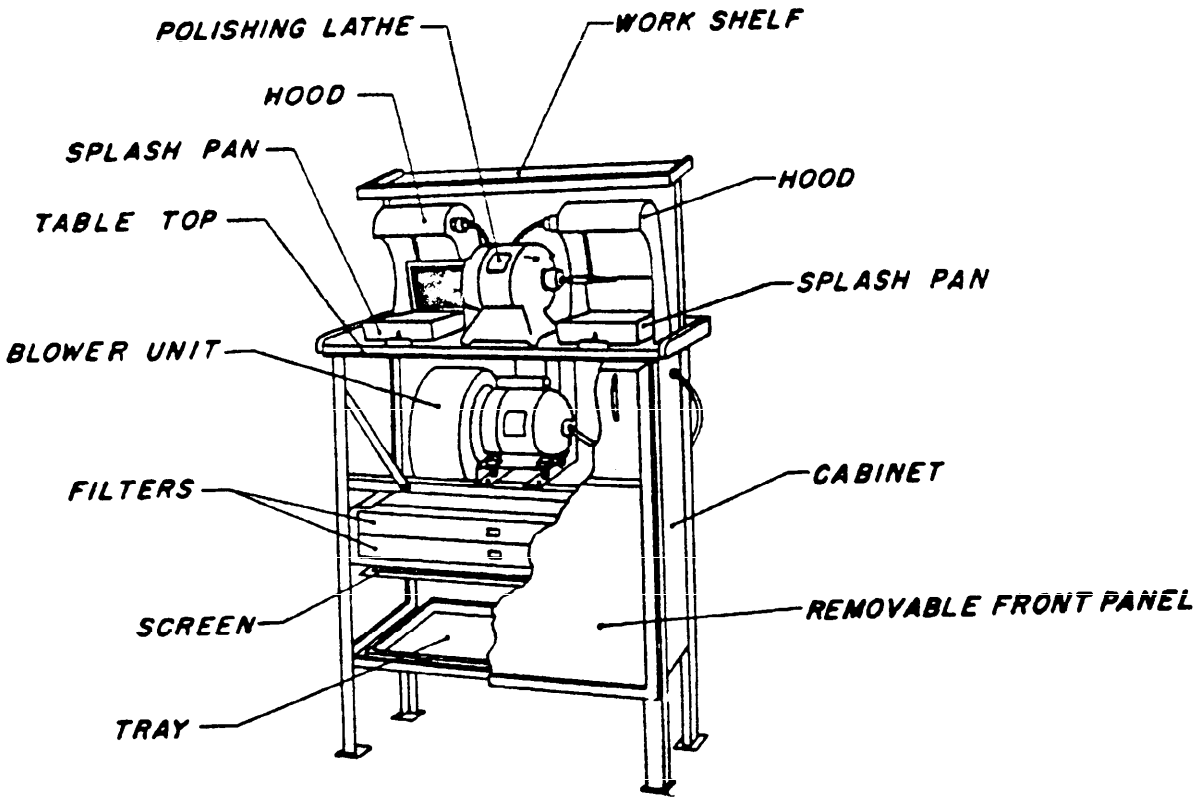


FIGURE 2.

MILITARY INTERESTS:

Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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